Tengsheng Zhang

ACADEMIC CURRICULUM VITAE

Postdoctoral Researcher, Fudan University
Visiting Scholar, Nanyang Technological University

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Academic Appointments

Sep 2026 - Present | Visiting Scholar
School of Materials Science and Engineering, Nanyang Technological University, Singapore.

Jun 2025 - Present | Postdoctoral Researcher
Advanced Materials Laboratory, Fudan University, Shanghai, China.

Education

Sep 2021 - Jun 2025 | Ph.D. in Physical Chemistry
Fudan University, Advanced Materials Laboratory, Shanghai, China.

Sep 2018 - Jun 2021 | M.S. in Materials Physics and Chemistry
Central South University, School of Materials Science and Engineering, Changsha, China.

Sep 2014 - Jun 2018 | B.Eng. in Materials Processing Engineering
Central South University, School of Materials Science and Engineering, Changsha, China.

Research Interests

Aqueous Batteries
Novel rechargeable battery chemistries, multielectron redox processes, and electrode-electrolyte interactions.

Redox Reactions
Electron-transfer processes, reaction pathways, and reaction reversibility in emerging energy storage systems.

Electrochemical Analysis
Electrochemical mechanisms investigated using in situ spectroscopy, synchrotron-based techniques, Raman spectroscopy, X-ray diffraction, and UV-Vis spectroscopy.

Research Experience

In situ characterization and electrochemical methods
Development of in situ experimental setups for synchrotron measurements, Raman spectroscopy, UV-Vis spectroscopy, X-ray diffraction, and infrared spectroscopy. Experience with synchrotron X-ray absorption, XRD, and SAXS measurements.

Materials characterization and data analysis
Experience with TOF-SIMS, XPS, SEM, Raman, FTIR, UV-Vis, and NMR measurements and data analysis.

Research Grants - Principal Investigator

2026 | Postdoctoral Innovative Talents Support Program
Regulation of Multielectron Reaction Pathways and Optimization of Selenium-Based Aqueous Batteries under Mesoporous Confinement. Grant No. BX2026085.

2025 | China Postdoctoral Science Foundation, 78th General Program
Development and Mechanistic Investigation of Selenium-Based Aqueous Batteries Involving Multielectron Reactions. Grant No. 2025M780904.

2025 | Shanghai Postdoctoral Domestic Dispatch Program
Investigation of Electrochemical Behavior and Failure Mechanisms of Flexible Aqueous Batteries.

2020 | Central South University Graduate Innovation Project
Two-Phase Mn-Based Composite Cathodes with High Cycling Stability for Aqueous Zinc-Ion Batteries and Their Electrochemical Performance. Grant No. 2020zzts443.

Research Grants - Participant

2024 | National Key R&D Program of China
Construction of High-Performance Aqueous Batteries and Key Preparation Technologies for Porous Materials. Grant No. 2024YFE0101100.

2020 | National Natural Science Foundation of China, Key Program
Development of New Rechargeable Battery Systems for Low-Cost, High-Specific-Energy, Long-Life Large-Scale Energy Storage. Grant No. 51932011.

Honors & Awards

• Gold Award, 2nd National Postdoctoral Innovation and Entrepreneurship Competition

• National Silver Award, 9th China International Internet+ College Students Innovation and Entrepreneurship Competition

• National Graduate Scholarship

• Xianghui Scholarship, Fudan University

• Outstanding Student, Fudan University

• HDR Best Presentation, International Conference on Materials for Green Future

English grant and award names are translations where no official English name was supplied. Award years are omitted because the source CV does not specify them.

Selected Publications

Author lists are abbreviated. Author roles follow the source CV.

1. Interfacial Lewis Acid Chemistry Enabled by Mesoporous MOFs Toward High-Performance Four-Electron Zinc-Iodine Batteries
F. Yang, T. Zhang, et al. *Angewandte Chemie International Edition (2026), e3627132.*
Co-corresponding author | 10.1002/anie.3627132

2. Engineering Solvent-Anion Interactions via Asymmetric Ethers for 200 Wh kg-1 Low-Temperature Anode-Free Sodium Pouch Cells
F. Huang, T. Zhang, et al. *Angewandte Chemie International Edition (2026), e3780082.*
Co-corresponding author | 10.1002/anie.3780082

3. High-Energy Aqueous Sulfur Battery Chemistry
X. Yu, T. Zhang, et al. *Advanced Materials (online 2025; volume 38, 2026).*
Co-first author | 10.1002/adma.202507561

4. Recyclable Biomacromolecular Hydrogel Electrolytes for Zn-Based Aqueous Batteries
T. Zhang, et al. *National Science Review 12, nwaf383 (2025).*
First author | 10.1093/nsr/nwaf383

5. Aqueous-S vs Organic-S Battery: Volmer-Step Involved Sulfur Reaction
T. Zhang, et al. *Journal of the American Chemical Society 147, 11501-11510 (2025).*
First author | 10.1021/jacs.5c01727

6. Protocol in Evaluating Capacity of Zn-Mn Aqueous Batteries: A Clue of pH
H. Yang, T. Zhang, et al. *Advanced Materials 35, 2300053 (2023).*
Co-first author | 10.1002/adma.202300053

7. Redox Mediator Chemistry Regulated Aqueous Batteries: Insights into Mechanisms and Prospects
T. Zhang, et al. *CCS Chemistry 4, 2874-2887 (2022).*
First author | 10.31635/ccschem.022.202202125

8. A Solid-to-Solid Metallic Conversion Electrochemistry Toward 91% Zinc Utilization for Sustainable Aqueous Batteries
Z. Hou, T. Zhang, et al. *Science Advances 8, eabp8960 (2022).*
Co-first author | 10.1126/sciadv.abp8960

9. Fundamentals and Perspectives in Developing Zinc-Ion Battery Electrolytes: A Comprehensive Review
T. Zhang, et al. *Energy & Environmental Science 13, 4625-4665 (2020).*
First author | 10.1039/D0EE02620D

10. Electrochemical Activation of Manganese-Based Cathode in Aqueous Zinc-Ion Electrolyte
T. Zhang, et al. *Advanced Functional Materials 30, 2002711 (2020).*
First author | 10.1002/adfm.202002711

Complete publication record: Google Scholar